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[GS PAPER II – GOVERNANCE • GS PAPER III – ECONOMY, ENVIRONMENT, AGRICULTURE] | **Publication Date: 07/10/2026**

Waste to Wealth: Turning Waste into Value



The article explores how waste can be transformed into valuable resources through composting, recycling, reuse and innovative waste-to-wealth initiatives. It highlights the circular economy, sustainable livelihoods and community participation in building cleaner, resource-efficient and environmentally sustainable communities.

Waste to Wealth: Turning Waste into Value

Syllabus areas:

Essay

GS Paper II – Governance

GS Paper III – Economy, Environment, Agriculture



Swachh Bharat Mission (SBM), launched in 2014, has evolved into a broader movement promoting cleanliness, waste management and behavioural change. Swachhata Hi Seva (SHS), launched in 2017, strengthened citizen participation. The **Waste to Wealth** approach treats waste as a resource, promoting circularity, resource efficiency and sustainable waste management. Under **SBM-U 2.0**, cities focus on source segregation, waste processing, recycling and value recovery.

A Compost Bin Brings the Change Home

In Vijayawada, senior citizen K. Subbalakshmi converts kitchen wet waste into compost for her garden, demonstrating sustainable household waste management. In Chandigarh, Kamaljeet Kaur Deol began home composting during the COVID-19 pandemic and later promoted awareness among citizens. Both examples show how simple household practices can support cleaner cities and sustainable waste management.



In Siddipet, Telangana, the ‘Swachh Badi’ waste management learning centre provides young people practical exposure to composting and responsible waste management. Similar centres across 141 municipal areas extend community learning. These initiatives show how simple household composting can reduce wet waste sent to landfills, support plant growth and promote sustainable waste management at its source.

Waste Hierarchy and Source Segregation

The initiatives reflect the **waste hierarchy**, which prioritises prevention, reduction, reuse, recycling and recovery, while treating disposal as the last option. Effective **source segregation** of wet, dry and other waste is essential because mixed and contaminated waste reduces recycling potential, resource recovery and the efficiency of waste-processing systems.



When a weed becomes an opportunity

In Borchila village of Assam’s Morigaon district, women are turning water hyacinth, locally called ‘pani meteka’, into useful handicrafts. The invasive aquatic plant rapidly spreads across rivers, ponds and other freshwater bodies, obstructing fishing and transportation. Supported by the Assam State Rural Livelihoods Mission, the initiative provides participating women with an additional income of at least Rs. 10,000 per month. By converting an environmental burden into marketable products, the initiative demonstrates how resource recovery can create livelihoods while promoting sustainable resource use. It also highlights the potential of community-led enterprises to address local environmental

challenges through innovative and productive solutions.

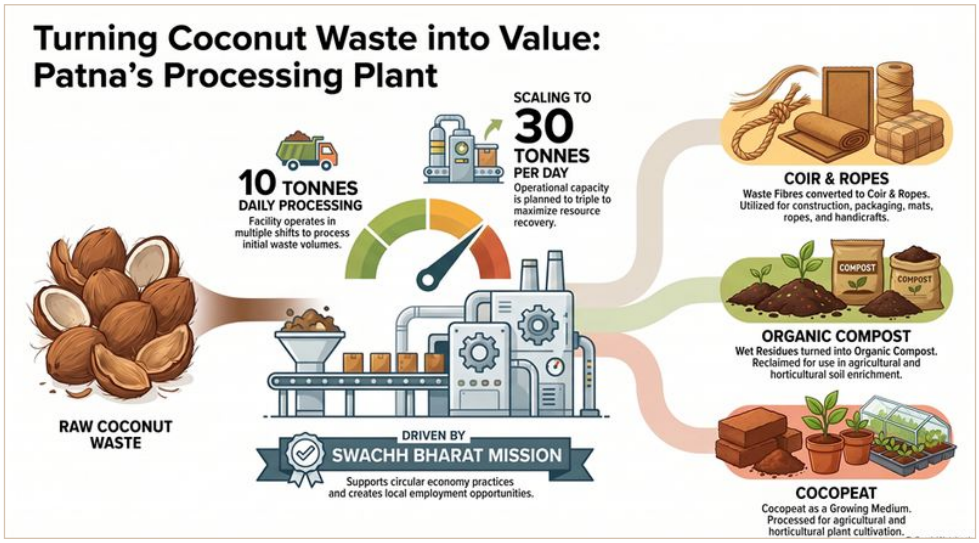
Coconut Waste Finds a New Market



In Patna, coconut shells, fibres and other residues are being converted into useful products instead of reaching landfills. The city generates around 25–30 metric tonnes of coconut shells daily. To manage this waste stream, the Patna Municipal Corporation has established a processing facility in Danapur, creating opportunities for resource recovery and productive reuse.

Coconut Waste Processing Plant in Patna

The initiative reflects the Swachh Bharat Mission's resource-based approach to waste management, converting coconut waste into useful products for agriculture, horticulture and commercial applications. The facility initially processes 10 tonnes daily and is designed to operate in multiple shifts. Its capacity is planned to increase gradually to 30 tonnes per day, supporting efficient resource recovery.



Ropes made from waste fibers:

Different coconut waste components can be recovered for productive use. Wet residues are converted into organic compost, while fibres are processed into coir for construction, packaging, mats and handicrafts. Cocopeat is useful in agriculture and horticulture as a growing medium. Such recovery reduces disposal, supports circular economy practices and creates livelihood opportunities.

Livelihoods and Green Economy

Waste recovery can generate **green jobs and local livelihoods**, particularly for women, Self-Help Groups, waste workers and small enterprises. Integrating community-based initiatives with formal municipal systems can make waste management environmentally sustainable while promoting employment, entrepreneurship, social inclusion and resource efficiency, contributing to an inclusive **green economy**.

When Waste Becomes a Public Space



The Waste to Wealth approach encourages cities to reuse discarded materials creatively. Waste to Wonder parks transform scrap into public attractions, reflecting circular economy principles of reducing, reusing, repairing and recycling materials for continued productive use.

Chandigarh's Rock Garden showcases creative reuse of industrial and household waste. Its 1.75-acre Waste to Wonder Park similarly transforms discarded iron sheets, rods, fans and automobile parts into public structures and attractions.

Delhi's Waste to Wonder Park repurposes industrial waste, scrap metal and discarded automobile parts into replicas and public structures. It also integrates solar trees and rooftop panels with 50 KW capacity. The generated electricity can support distribution companies, creating revenue while combining waste recovery, urban beautification, renewable energy and sustainable resource use.

From Waste to Resource

The Waste to Wealth approach transforms discarded materials into useful resources through segregation, recovery and processing. Waste can become components of public spaces, creative attractions and economic assets. By recognising its resource potential, this approach reduces disposal, promotes cleaner communities, improves resource efficiency and supports India's transition towards a circular economy.

Waste to Wealth refers to recovering, recycling, reusing and processing discarded materials to generate economic, social and environmental value, thereby reducing waste disposal and promoting a circular economy.



Waste-to-Wealth: Sectoral Opportunities

Waste Stream	Potential Resource
Wet waste	Compost, biogas

Waste Stream	Potential Resource
Coconut waste	Coir, cocopeat, compost
Plastic	Recycled products
E-waste	Metals and components
Construction waste	Aggregates
Agricultural waste	Bioenergy, biochar
Sewage	Treated water, biogas
Used batteries	Critical minerals/materials

Why Waste to Wealth Matters for India



Environmental benefits

- Reduces landfill pressure
- Prevents soil and groundwater contamination
- Reduces methane emissions
- Conserves natural resources
- Promotes recycling and resource recovery

Economic benefits

- Creates green jobs
- Generates livelihoods
- Reduces dependence on virgin raw materials
- Creates local enterprises
- Converts waste streams into economic value

Social benefits

- Supports informal waste workers
- Creates opportunities for SHGs and women
- Promotes community participation
- Improves public health and sanitation